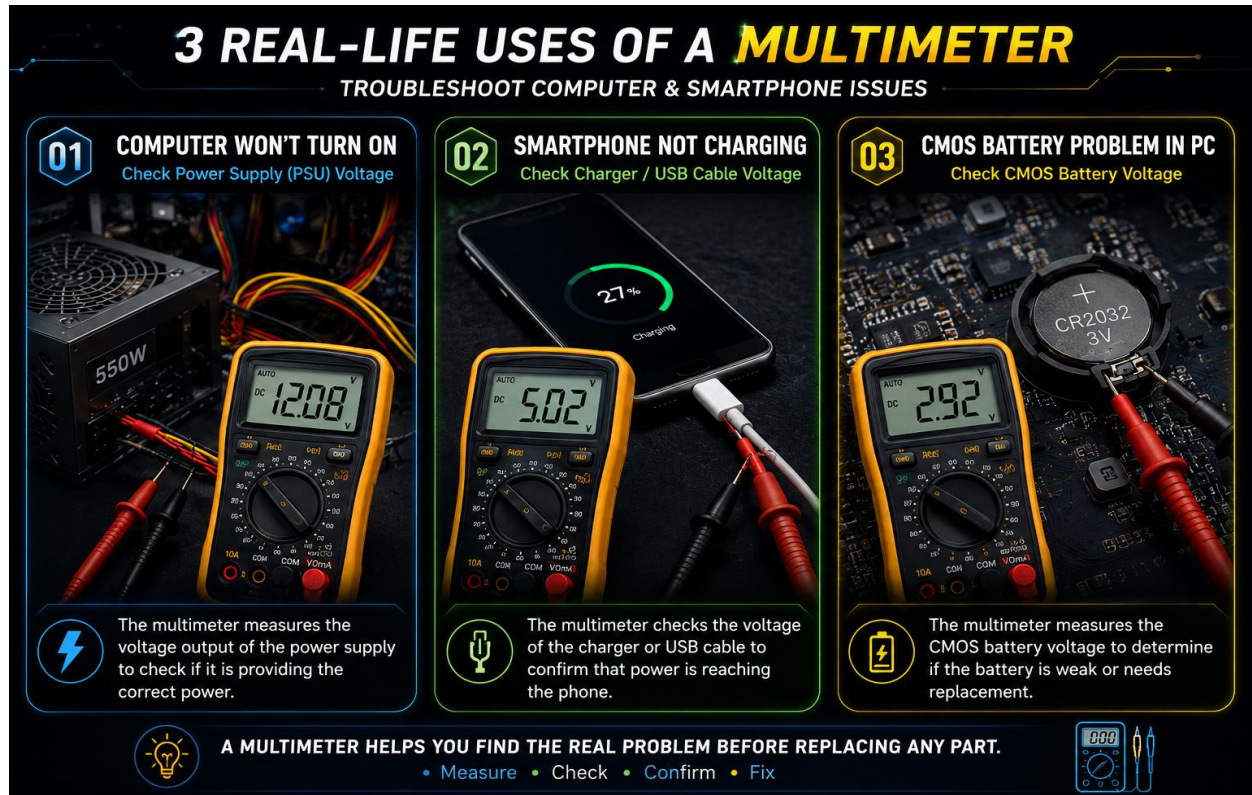


1 List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.



1. Computer won't turn on

1. Measure the PSU (Power Supply Unit) voltage.
2. Checks whether the computer is receiving correct power.

2. Smartphone not charging

- Measure the charger or USB cable voltage.
- Confirms that power is reaching the phone.

3. CMOS battery problem in PC

- Measure the CMOS battery voltage.
- Identifies if the battery is weak and needs replacement.

2 Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

❖ POST CARD ::



- I watched a demo video of a POST Card used on a motherboard.
- The POST Card shows the hardware checking process done by the BIOS.
- It displays Hexadecimal Debug Codes such as C1, 55, and 80 on the LED screen.
- Each code shows a different hardware testing stage.
- In the demo, the final code was 80, which means the POST process finished successfully.

- These codes help identify RAM or motherboard boot problems.
- Therefore, a POST Card is a useful diagnostic tool for finding computer hardware faults.

3 Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.



The screenshot shows the HWMonitor application window. The title bar is 'HWMonitor' with standard window controls. The menu bar includes 'File', 'View', 'Tools', and 'Help'. The main content area is a table with columns 'Sensor', 'Value', 'Min', and 'Max'. The sensors are categorized into AMD Radeon(TM) Graphics, 0x35,0x33,0x32,0x4E,0x00,0x00,..., VMware Virtual Ethernet Adap..., Npcap Loopback Adapter, and Realtek 8821CE Wireless LAN ...

Sensor	Value	Min	Max
AMD Radeon(TM) Graphics			
Clocks			
Graphics	400.0 MHz	400.0 MHz	675.0 MHz
Memory	400.0 MHz	400.0 MHz	1200.0 MHz
Utilization			
GPU	5.0 %	0.0 %	26.0 %
Video Engine	0.0 %	0.0 %	0.0 %
3D	6.1 %	0.0 %	29.2 %
Copy	0.4 %	0.0 %	6.4 %
Compute	0.0 %	0.0 %	0.0 %
Dedicated Memory	428.7 MB	403.5 MB	432.5 MB
Shared Memory	193.8 MB	187.8 MB	229.7 MB
Memory	86.5 %	81.4 %	87.3 %
Speed			
PCIe Link Speed	8.00 GT/s	8.00 GT/s	8.00 GT/s
0x35,0x33,0x32,0x4E,0x00,0x00,...			
Voltages			
Current Voltage	11.956 V	11.905 V	12.024 V
Powers			
Charge Rate	-7.74 W	-7.74 W	-6.14 W
Capacities			
Designed Capacity	40995 mWh	40995 mWh	40995 mWh
Full Charge Capacity	28283 mWh	28283 mWh	28283 mWh
Current Capacity	26426 mWh	26426 mWh	26539 mWh
Levels			
Wear Level	31.0 %	31.0 %	31.0 %
Charge Level	93.4 %	93.4 %	93.8 %
VMware Virtual Ethernet Adap...			
Speed	Download Speed	0.00 Mbps	0.00 Mbps
	Upload Speed	0.00 Mbps	0.00 Mbps
VMware Virtual Ethernet Adap...			
Speed	Download Speed	0.00 Mbps	0.00 Mbps
	Upload Speed	0.00 Mbps	0.00 Mbps
Npcap Loopback Adapter			
Speed	Download Speed	0.00 Mbps	0.00 Mbps
	Upload Speed	0.00 Mbps	0.00 Mbps
Realtek 8821CE Wireless LAN ...			
Speed	Download Speed	0.07 Mbps	0.82 Mbps
	Upload Speed	0.02 Mbps	0.23 Mbps

Ready

The screenshot shows the HWMonitor application window with a dark theme. The window title is 'HWMonitor' and it has a menu bar with 'File', 'View', 'Tools', and 'Help'. The main content area displays a tree view of system sensors. The 'DESKTOP-XXXXXX' section is expanded, showing 'Dell Inc. 0C3WG4' and 'AMD Ryzen 5 Mobile 5500U'. The 'AMD Ryzen 5 Mobile 5500U' section is further expanded, showing various sub-sections like Voltages, Temperatures, Powers, Currents, Levels, Clocks, and Utilization. The 'KBG50ZNS512G NVMe KIOXIA' section is also expanded, showing Temperatures, Counters, and Utilization. The data is presented in a table with columns for Sensor, Value, Min, and Max.

Sensor	Value	Min	Max
DESKTOP-XXXXXX			
Dell Inc. 0C3WG4			
Temperatures			
TZ.TZ01	53.0 °C	50.0 °C	54.0 °C
Utilization			
Physical Memory Load	91.0 %	89.0 %	95.0 %
Available Physical Memory	0.7 GB	0.4 GB	0.8 GB
Virtual Memory Load	60.5 %	60.0 %	61.5 %
Available Virtual Memory	8.2 GB	8.0 GB	8.3 GB
AMD Ryzen 5 Mobile 5500U			
Voltages			
VID (Max)	0.731 V	0.700 V	0.750 V
CPU VDD	0.744 V	0.694 V	0.750 V
SoC VDD	0.663 V	0.663 V	0.813 V
Temperatures			
Cores (Max)	52.4 °C	50.9 °C	53.8 °C
Package	53.3 °C	50.8 °C	54.5 °C
L3 Cache #0	52.2 °C	50.7 °C	53.0 °C
L3 Cache #1	51.8 °C	50.3 °C	52.6 °C
Powers			
Package	6.11 W	2.11 W	7.68 W
Cores	0.55 W	0.26 W	3.10 W
PPT Limit	30.00 W	30.00 W	30.00 W
Currents			
CPU VDD	6.24 A	0.00 A	17.67 A
SoC VDD	8.31 A	4.16 A	15.59 A
Levels			
Energy Performance	0.0 %	0.0 %	60.0 %
Clocks			
Cores (Max)	1397.0 MHz	1396.6 MHz	2071.5 MHz
CPU BCLK	99.8 MHz	99.7 MHz	99.9 MHz
Memory Controller	798.3 MHz	797.8 MHz	1598.0 MHz
Memory	1596.3 MHz	1595.9 MHz	1597.7 MHz
Utilization			
Processor	13.8 %	5.8 %	55.7 %
CCX #0	25.4 %	0.0 %	60.6 %
CCX #1	2.3 %	0.0 %	51.9 %
KBG50ZNS512G NVMe KIOXIA			
Temperatures			
Assembly	41.0 °C	41.0 °C	42.0 °C
Sensor 1	41.0 °C	41.0 °C	42.0 °C
Counters			
Power-On-Hours	975 hrs	975 hrs	975 hrs
Power Cycle Count	8257	8257	8257
Health Status	97 %	97 %	97 %
Utilization			

- I downloaded and installed **HWMonitor**, a free software diagnostic tool, on my computer.
- I opened **HWMonitor** and checked the main dashboard.

- It showed important system health details such as **CPU temperature, CPU usage, voltage, power, GPU information, storage temperature, and SSD health status.**
- I checked the **Value, Minimum (Min), and Maximum (Max)** readings of different hardware components.
- I took screenshots of the HWMonitor dashboard showing these system details.
- Before submitting the screenshots, I **cropped or blurred any personal information** for privacy and safety.

4 Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it Checks	Example Problem Detected
Multimeter	Checks voltage, current, and resistance.	Low power supply voltage.
POST Card	Checks hardware during computer startup.	RAM or motherboard problem.
Software Diagnostics	Checks system and hardware health.	High CPU temperature or SSD health problem.